

Intrinsic Causal Geometry: Treatment Effect Estimation in Non-Transportable Systems Under the Lucas Critique

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Abstract

This paper presents a framework for extending causal inference methods to handle treatment effect heterogeneity encoded in covariance structure, while identifying vulnerabilities in current approaches that warrant careful attention. I argue that three challenges affect causal inference in adaptive systems: (1) **transportability**—verifying that mechanisms hold in deployment contexts is difficult; (2) the **Lucas Critique**—publishing causal findings can change the systems being studied; and (3) **latent complexity**—treatment effect modifiers often span dimensions that exceed practical modeling capacity.

I trace the evolution of my thinking through two methodological frameworks. First, **AutoITE** (Automated Individual Treatment Effect Estimation), which used residual-based matching to discover latent environments. While AutoITE achieved strong results— $r = -0.94$ correlation with hidden confounders where Causal Forests achieve $r = 0.00$ —I ultimately moved away from this framework. The T-Learner with Ridge regression achieved comparable performance, but the deeper realization was that the concept of “latent environments” suggests exclusive pockets of meaning. In reality, I likely face **countless latent spaces of sparse membership**. My initial view of discrete environments was flawed, and this shift in perspective—rather than performance metrics alone—necessitated abandoning the environment discovery framework.

This motivated **Intrinsic Causal Geometry (ICG)**, which shifts from environment discovery to **manifold matching**: finding similar units based on the geometric structure of their covariance matrices rather than membership in unobservable categories. The key insight is that *distance error is far less consequential than membership error*—being slightly wrong about position on a continuous manifold produces bounded error, while misclassifying discrete latent type produces discontinuous harm.

The **Unified ICG** approach—combining Euclidean distance on means (Levels) with Riemannian distance on covariances (Structure)—achieves 72.5% lower total MAE than Random Forest across stress tests designed to expose different failure modes. ICG captures second-order heterogeneity (correlation patterns) that feature-based methods miss, while acknowledging its own limitation to second-order structure.

I extend Lucas’ Critique to show that it operates through **gradual decay**, not discrete collapse, with mechanism stability varying by domain (physics most stable, behavioral systems least). I argue that treatment should be modeled as a **vector or matrix**, not a scalar, and propose **closed-loop coupled interventions** that verify state changes rather than assuming mechanism stability.

This paper aims to complement, not replace, existing causal inference methods by extending their reach to covariance structure while promoting appropriate epistemic humility about the limits of causal knowledge in adaptive systems.

Keywords: Causal Inference, Riemannian Geometry, Manifold Learning, Treatment Effect Heterogeneity, Lucas Critique, Transportability, Covariance Structure

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Part I

The Fundamental Problem of Causal Inference

1 Introduction: Fundamental Challenges in Causal Inference

1.1 The Promise and the Challenge

Treatment effect heterogeneity—the reality that identical treatments produce different outcomes for different individuals—is central to personalized medicine, education, and policy. Current state-of-the-art methods (Causal Forests (Wager and Athey, 2018), X-Learners (Künzel et al., 2019), Double Machine Learning (Chernozhukov et al., 2018)) estimate Conditional Average Treatment Effects (CATE):

$$\tau(X) = \mathbb{E}[Y(1) - Y(0) \mid X] \quad (1)$$

where $Y(1)$ and $Y(0)$ are potential outcomes under treatment and control, and X represents observed covariates.

These methods share a common assumption: **all causally relevant heterogeneity is explainable by observed features X** . They partition the covariate space (via splits, weights, or kernels) and assume that individuals within the same partition share similar treatment effects.

My analysis suggests this assumption faces three key challenges:

1. **Transportability Verification:** The assumption that causal mechanisms estimated from training data will hold in deployment contexts is difficult to test prospectively. I can often only verify it after the fact, by which point treatment decisions have been made.
2. **Lucas’ Critique:** The act of discovering and publishing a causal relationship can change the system being studied. “Treated vs. untreated” becomes “treated vs. treatment withheld” the moment findings are actualized.
3. **Latent Complexity:** The true space of treatment effect modifiers often extends to dimensions that exceed practical modeling capacity.

1.2 Conceptual Origins and Structure

The approach detailed in this paper stems from a specific conceptual derivation regarding the nature of samples and treatments.

I considered that for vectors individually, if I calculate an inverse matrix, I am essentially encoding or finding the work required to “undo” that sample or vector. If I can represent samples as matrices or intrinsic shapes—where there is no global origin and axes are constructed from the intrinsic shapes themselves—and with this view find the samples that require the minimum effort to align, I can find samples that group together by structural effects.

Furthermore, I considered that I should view treatment as a vector or matrix, because the correct view is that latent space transforms relationships, including those of treatment. If I view treatment as a scalar, this becomes infeasible. Moreover, the coupling of treatments becomes feasible—such as the proposal to encourage a patient to lose weight and gain muscle mass to see if the chances of success for a primary treatment can be improved. This otherwise cannot be done with two separate models independently, as is noted by extensions of the Lucas critique.

The framework presented here maps this logic onto existing implementations. I did not initially derive this from physics; rather, I observed the parallel after the fact and use it here as a defense for the additional complexity of the approach. The connection to physics suggests that the complexity is not arbitrary, but necessary to capture the invariant properties of the system.

Part I establishes the fundamental challenges, arguing that causal inference faces limitations rooted in the nature of knowledge and reflexivity.

Part II presents AutoITE, my first attempt to address these problems through residual-based discovery.

Part III develops Intrinsic Causal Geometry (ICG), which implements the “minimum effort to align” concept via manifold matching.

Part IV connects causal inference to physics, using the parallel to defend the complexity required to model adaptive systems.

Part V extends Lucas’ Critique to its full implications.

Part VI presents empirical validation through stress tests.

Part VII concludes with implications for practice.

1.3 The Ethical Imperative: Selective Empathy and the Culling of Divergence

My previous work on AutoITE, now archived as a “tombstone,” highlighted a grim truth that motivates this entire endeavor: my reliance on averages—whether ATE, CATE, or CQTE—systematically culls and penalizes divergence.

Divergence can occur on any dimension. When I rely on group averages, I exercise **selective empathy**: I protect specific groups that I have explicitly defined and measured, while ignoring and penalizing individuals who diverge on dimensions I failed to model. I accept high error rates for “outliers” as the cost of model stability.

The solution is to protect not groups, but **individuals**. By pursuing Individual Treatment Effects (ITE) through geometric matching, I refuse to aggregate unique divergences into convenient buckets. I aim to protect the anomaly as zealously as the average.

2 The Unstable Foundation: Transportability

2.1 What Transportability Requires

All causal inference methods assume some form of invariance: that the causal mechanisms estimated from training data will hold in the deployment context. This is the *transportability* assumption (Pearl and Bareinboim, 2014).

Formally, transportability requires:

$$P^*(Y \mid do(T), X) = P(Y \mid do(T), X) \quad (2)$$

where P is the distribution in the training domain and P^* is the distribution in the target domain. The assumption is that the *interventional* distribution—the distribution of outcomes under intervention—is identical across contexts.

2.2 Why Transportability Cannot Be Tested

The fundamental problem is that transportability **cannot be tested**. I can only verify it after the fact, by observing whether predictions hold in the new context. But by then, treatment decisions have already been made.

Consider the epistemological structure:

1. I estimate $\hat{\tau}(X)$ from training data.
2. I deploy treatment decisions based on $\hat{\tau}(X)$.
3. I observe outcomes in the deployment context.
4. Only *then* can I assess whether transportability held.

The harm has already occurred by step 4. There is no way to verify transportability *before* I make consequential decisions.

But the problem runs deeper. Even successful after-the-fact verification provides no guarantee of future stability. Under the gradual decay framework introduced in Part V, validity decays according to:

$$\text{Validity}(t) = \text{Validity}_0 \cdot e^{-\lambda \cdot D(t)} \quad (3)$$

where $D(t)$ measures cumulative dissemination. A relationship that appears to transport successfully may simply be in the early phase of decay—the half-life may exceed the verification window. Worse, the act of publishing verification results itself contributes to $D(t)$, potentially accelerating the very decay I failed to detect.

This creates an epistemological trap: *evidence of invariance is not proof of invariance*. A mechanism that has held across ten contexts may fail in the eleventh—not because of sampling error, but because the publication of its success in ten contexts changed the system. The recursive nature of Lucas’ Critique means that successful verification can be self-undermining.

2.3 Where Invariance Actually Holds

The only systems where I can *guarantee* invariance are:

1. **Physical laws:** The relationship $F = ma$ holds regardless of who measures it or when. The gravitational constant does not change because physicists discovered it. Quantum mechanics does not adapt to observation (the measurement problem notwithstanding, the *laws* are invariant).
2. **Random mutations in game theory:** Evolutionary dynamics with random variation operate by mechanisms that don’t depend on observers. A mutation that confers fitness advantage will spread regardless of whether biologists have characterized it.

Systems studied in economics and social science clearly do not fall into these categories—their mechanisms respond rapidly to observation and the dissemination of knowledge. Medicine occupies an intermediate position: the underlying pharmacology may be stable, but clinical practice is subject to gradual decay as knowledge spreads.

2.4 Invariance in Physical Systems

While most physical systems and their governing laws are observer-independent—their behavior remains unaffected by human knowledge or observation—this stands in stark contrast to adaptive systems found in social science, economics, and medicine, which often respond to knowledge dissemination. A profound parallel, however, exists within physics itself: quantum mechanics. Here, observation inherently disturbs the observed, and the act of measurement affects the system’s state—a physicist’s version of Lucas’ Critique. My Intrinsic Causal Geometry framework mirrors this aspect of quantum mechanics; it attempts to operate like ‘weak measurement,’ seeking to extract similarity information from continuous manifolds while minimizing the ‘collapse’ of heterogeneity that occurs with discrete classifications.

3 The Principle of Proxy Instability

3.1 Limitations of Static Feature Maps

I formalize the core insight that motivates this work:

Definition 1 (Latent Environment). A *latent environment* U is an unobserved variable that causally influences both baseline outcomes and treatment effects. Formally, for individual i :

$$Y_i = f(X_i, T_i, U_i, \epsilon_i) \quad (4)$$

where ϵ_i is random noise independent of (X, T, U) .

Critical clarification: A latent environment is not a physical or administrative label such as “Hospital A” or “Hospital B.” Two patients from different hospitals may share the same latent environment (e.g., similar metabolic dysfunction), while two patients from the same hospital may have entirely different latent environments. The latent environment captures the individual’s underlying causal state—their genetic predisposition, physiological subtype, or behavioral pattern—not their membership in an observable group.

Assumption 1 (Proxy Instability). The mapping between observed features X and latent environment U is *non-stationary*:

$$P(U_i | X_i) \neq P(U_j | X_j) \quad \text{for } i \sim \mathcal{D}_{\text{train}}, j \sim \mathcal{D}_{\text{test}} \quad (5)$$

where $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$ may differ by time, domain, or population.

3.2 Examples of Proxy Instability

Consider the following:

- In 1990, “high income” proxied for “status-seeking behavior” in consumer studies. By 2025, this relationship has inverted in many demographics.
- Pre-treatment glucose levels may proxy for “metabolic dysfunction” in one population, but “athletic carbohydrate loading” in another.
- “High test scores” may indicate “intrinsic motivation” for some students, but “parental pressure” for others.

The correlation between X (observable features) and U (latent causal state) is **unstable**. Therefore, models that lock in static feature-to-subgroup mappings fail when:

1. The proxy relationship shifts over time (distribution shift)
2. The proxy relationship is incomplete (unobserved confounding)
3. The proxy relationship is orthogonal ($U \perp X$)

3.3 Limitations of Static Feature Mapping

Because the proxy mapping is unstable, learning a fixed function $h : \mathcal{X} \rightarrow \mathcal{U}$ that generalizes across contexts becomes difficult. Standard CATE methods, which condition only on X , may underperform when proxy instability is present.

This observation suggests an alternative framing: rather than “learn a function from features to effects,” consider “find similar cases at prediction time using dynamic information.” This reframing motivates the geometric matching approach developed in later sections.

4 The Irreducible Complexity of Latent Confounding

4.1 The Dimensionality of Reality

For any treatment decision, the true set of relevant latent factors includes:

- Genetic variants affecting drug metabolism
- Microbiome composition affecting absorption
- Sleep quality on the days surrounding treatment
- Stress levels and cortisol dynamics
- Social support availability
- Trust in the healthcare system
- Prior experiences with similar treatments
- Concurrent medications and supplements
- Dietary patterns in recent weeks
- Environmental exposures (pollution, allergens, temperature)
- Circadian timing of treatment administration
- Psychological state (depression, anxiety, optimism)
- Economic stressors affecting adherence
- Family dynamics and caregiving responsibilities

But even this list drastically understates the problem. Consider just a few examples in depth:

4.1.1 Bloodflow and Cardiovascular State

Treatment absorption and distribution depend on cardiac output, blood viscosity, vascular tone, and regional perfusion—all of which vary by time of day, recent activity, hydration status, temperature, and emotional state. I measure blood pressure; I do not measure the microcirculatory dynamics that actually determine drug delivery to target tissues.

4.1.2 Socioeconomic Proxies

What I call “socioeconomic status” is a crude projection of a hundred-dimensional reality. Income correlates with nutrition, but through pathways mediated by food deserts, cultural food preferences, time available for cooking, knowledge of nutrition, psychological relationship with food, childhood eating patterns, and social eating norms. Each of these is itself a proxy for deeper variables. The DAG would have hundreds of nodes, most unmeasured.

4.1.3 Water Quality and Environmental Exposure

A patient’s residential water supply affects mineral intake, microbiome composition, toxin exposure, and hydration behavior. But “water quality” varies by building age, pipe material, local source contamination, filtration practices, and consumption patterns. Two patients in the same city may have radically different exposures, unmeasured and unmeasurable.

4.1.4 Location as Confound

Geographic location encodes air quality, altitude (affecting oxygen delivery), local pathogen ecology, UV exposure, healthcare access, social capital, and hundreds of other factors. I record “ZIP code”; I do not record the causal mechanisms it represents.

4.1.5 Introversion and Hospital Environment

Introverted patients experience hospital stays differently than extroverts—more stress from lack of privacy, different responses to group therapy, different relationships with rotating staff. Personality is sometimes assessed; its interaction with the hospital environment is not.

4.1.6 Hospital Micro-Environment

Even within a single hospital, outcomes differ by unit, shift, staff fatigue levels, concurrent patient load, and the mood of the care team. A patient admitted on a Monday morning receives different care than one admitted Friday night.

4.2 From Explicit Modeling to Geometric Matching

This list could extend to a hundred or more dimensions, each with complex interactions. Explicitly modeling this full structure from observed data exceeds practical capacity in most applications.

But here is the key insight: I do not need to model the infinite confounders explicitly. I can *match* based on observable structure through **Intrinsic Geometry**.

The covariance matrix Σ of observed variables serves as a **second-order structural proxy** for the latent state. It captures how measured variables co-vary—reflecting the influence of latent factors that affect multiple observables through their pairwise interactions.

An important caveat: Σ is *not* a sufficient statistic except under Gaussian assumptions. Biological data often exhibits higher-order structure—skewness, kurtosis, and complex tail behavior—that covariance alone cannot capture. What Σ *does* capture is the **interaction structure**: the second-order dependencies that drive phenomena like the “Interaction Gate” where correlation patterns determine treatment effects. For heterogeneity encoded in higher moments, ICG would require extension to higher-order tensors.

This is the resolution: *I cannot model U , but I can match units whose second-order observable structure suggests similar U .* The geometric structure of the data becomes an approximate proxy for the unknowable latent state—powerful for interaction-driven heterogeneity, limited for distributional heterogeneity beyond the second moment.

Part II

AutoITE: The First Attempt

5 The Residual-Based Insight

5.1 From Features to Behavior

If the mapping $X \rightarrow U$ is non-stationary, one cannot learn a global function that reliably recovers U from X . Instead, one must perform **Just-in-Time (JIT) discovery**: at prediction time, using instance-specific information—not just the static feature vector X_i , but the dynamic residual pattern R_i —to identify the latent state U_i .

This is the core insight of AutoITE: **residuals are more stable proxies than features**. While the relationship “Income $\rightarrow$ Status” may shift, the relationship “Baseline residual $\rightarrow$ Latent metabolic state” is grounded in a fundamental coupling: latent environments that affect treatment response also affect baseline outcomes, leaving a consistent fingerprint.

5.2 The Baseline Coupling Assumption

The key enabling assumption for AutoITE is **baseline coupling**: latent environments that affect treatment response must also affect baseline outcomes.

Assumption 2 (Baseline Coupling). *Latent environment U affects baseline outcomes Y_0 :*

$$Y_0 = f(X) + g(U) + \epsilon \quad (6)$$

where $g(U)$ is a non-constant function. This implies that baseline residuals $R = Y_0 - \hat{f}(X)$ capture information about U .

Intuition: If high genetic risk makes you both (a) more likely to have elevated baseline biomarkers and (b) more susceptible to treatment side effects, then your baseline residual (deviation from expected biomarkers given demographics) is a fingerprint of your genetic risk.

When Coupling Fails: If U affects only treatment effect but not baseline (“interaction-only confounders”), AutoITE cannot recover U from baseline residuals.

6 The AutoITE Architecture

6.1 Four-Stage Pipeline

AutoITE operates in four stages:

6.1.1 Stage 1: Global Anchor (The Naive Prior)

Train a deliberately simple Ridge regression on observed features:

$$\hat{Y}_{global} = \arg \min_f \sum_{i=1}^n (Y_i - f(X_i, T_i))^2 + \alpha_{global} \|f\|^2 \quad (7)$$

Why naive? A complex non-linear model would explain away the variance. A simple linear model with $\alpha_{global} = 1.0$ forces the latent signal into the residual.

6.1.2 Stage 2: Residual Embedding (The Latent Fingerprint)

Compute baseline residuals as dynamic proxies for U :

$$R_i = Y_{0,i} - \hat{f}_{global}(X_i, T = 0) \quad (8)$$

Under baseline coupling, $R_i \approx g(U_i) + \epsilon_i$. Individuals with similar residual patterns $R_i \approx R_j$ likely share similar latent states $U_i \approx U_j$, **even if their feature vectors $X_i \neq X_j$ differ**.

This is the key distinction from feature-based methods: clustering by **behavior** (how you deviate from expectation) rather than **appearance** (what your demographics are).

6.1.3 Stage 3: Just-in-Time Local Learning (The Last Meter)

For each test individual i , find k nearest neighbors in residual space:

$$\mathcal{N}_i = \{j : j \in \text{top-}k \text{ by } |R_j - R_i|\} \quad (9)$$

Train a local Ridge regression on this neighborhood:

$$\hat{Y}_{local}^{(i)} = \arg \min_f \sum_{j \in \mathcal{N}_i} (Y_j - f(X_j, T_j))^2 + \alpha_{local} \|f\|^2 \quad (10)$$

with $\alpha_{local} = 0.01$ (much smaller than global) to allow fitting strong local effects.

Metaphor: The global model flies you to the right neighborhood (based on demographics). The local model lands on the specific house (based on residual fingerprint).

6.1.4 Stage 4: Density Fusion (Adaptive Weighting)

Blend global and local predictions:

$$\hat{\tau}_i = \lambda_i \cdot \hat{\tau}_{local}^{(i)} + (1 - \lambda_i) \cdot \hat{\tau}_{global} \quad (11)$$

where $\lambda_i = \frac{1}{1 + \text{median}(d_{ij})}$ and d_{ij} are distances to neighbors in residual space.

7 AutoITE: Empirical Success

7.1 Orthogonality Recovery

Core Finding: When $U \perp X$ (features provide zero information about latent state), Causal Forest achieves **zero correlation** with the latent confounder, while AutoITE achieves $r = -0.94$ via residuals.

Table 1: Orthogonality Recovery: Feature-Based Methods Underperform Under Proxy Instability

Method	Corr($\hat{\tau}$, U)	Detection Rate	MAE	Max
Global Ridge (no baseline)	-0.02	28.1%	0.748	2.5
Causal Forest (features only)	0.00	27.3%	0.230	2.5
XGBoost S-Learner	-0.01	26.8%	0.260	2.7
X-Learner	0.00	27.1%	0.245	2.4
AutoITE ($k = 1000$)	-0.94	97.5%	0.095	2.5

Interpretation:

- All feature-based methods perform at random chance ($\sim 27\%$ detection vs 50% base rate) because features provide zero information about U under orthogonality.
- AutoITE achieves 97.5% detection by using baseline residuals as dynamic proxies, recovering $\pi_{lat}(U)$ with 59% lower MAE than Causal Forest.
- **Critical observation:** All methods share a maximum error of ~ 2.5 . This is a fundamental identifiability limit, not an algorithm limitation.

7.2 Safety Lift

In a simulated hidden toxicity trial where treatment is beneficial on average but lethal for a latent subgroup:

- AutoITE achieves 59% better MAE than Causal Forest (0.095 vs 0.230)
- With 15% triage, MAE reduces to 0.042 (82% better than Causal Forest)
- Deaths reduced from 8 to 3 (62% reduction) at 80% automated coverage

7.3 The 356-Death Difference

When comparing scenarios with and without baseline coupling:

Scenario	Deaths	Detection Rate
U affects only τ (no baseline signal)	379	62% (random)
U affects baseline + τ (coupling)	23	98%
Difference	356 fewer	+36 percentage points

This $16\times$ difference demonstrates the catastrophic consequence of interaction-only confounders and the value of baseline coupling when it exists.

8 AutoITE: The Limitations Discovered

8.1 Comparison with Simpler Methods

Before developing the theoretical framework further, I must acknowledge an empirical finding that informs the subsequent development: **a simple T-Learner with Ridge regression achieves comparable or better performance than AutoITE.**

The T-Learner approach—training separate Ridge models on treated and control groups—matches AutoITE’s performance with dramatically less complexity. The four-stage AutoITE architecture (global anchoring, residual embedding, JIT local learning, density fusion) was designed to capture latent heterogeneity that simpler methods might miss. Yet the simpler method performs equally well.

This is not a failure of the philosophical insight that residuals contain latent information. It is a lesson about the appropriate level of abstraction. The insight suggests that the value lies not in complex architectures but in the matching principle itself—which can be implemented simply.

8.2 The Discrete Environment Problem

AutoITE’s framing assumed I could “discover” discrete latent environments—categories like “high-risk” vs. “low-risk,” “responder” vs. “non-responder.” I now recognize this framing, while intuitive, may have been overly ambitious for the complexity of real systems.

The problem is not that residuals fail to contain information about latent structure—they demonstrably do. The problem is that the concept of discrete “latent environments” is itself too crude to capture reality.

Consider what I am really asking: can I partition patients into membership categories based on unobserved characteristics? The answer is no, for three reasons:

1. **Latent spaces are orthogonally multidimensional:** There is no single latent variable U that determines treatment response. Instead, there exist *many* latent dimensions—genetic predisposition, metabolic state, psychological resilience, social support, environmental exposures—each of which may modulate the relationship between X , T , and Y in different and orthogonal ways.
2. **Membership error is catastrophic:** Discrete clustering requires hard boundaries. A patient just inside the “high-risk” boundary is treated identically to one deep inside it, while being treated completely differently from a patient just outside. This “membership error” produces discontinuous harm.
3. **Distance error is bounded:** In contrast, if I view the latent space as a continuous manifold and estimate treatment effects locally based on geometric distance, being slightly wrong about a patient’s position produces proportionally small error. The harm is continuous and bounded.

This insight—that *distance error is far less consequential than membership error*—motivates the shift from latent environment discovery to manifold matching.

Part III

Intrinsic Causal Geometry

9 From Discovery to Matching

9.1 The Philosophical Shift

Intrinsic Causal Geometry (ICG) abandons the goal of latent environment discovery. Instead, it adopts a purely geometric approach:

I do not ask “what latent type is this patient?” I ask “which historical patients are geometrically similar to this one?”

The key insight is that treatment effect heterogeneity is encoded in the **structure of relationships** between covariates, not in the covariates themselves. Two patients with identical observed features X may have completely different treatment responses if the *correlation structure* among their measurements differs.

9.2 Why Structure Matters

Consider two patients with identical mean blood pressure, heart rate, and glucose levels. If Patient A shows high correlation between blood pressure and heart rate (suggesting autonomic coupling), while Patient B shows zero correlation (suggesting decoupling), they may respond very differently to cardiovascular interventions—even though their summary statistics are identical.

Traditional methods see identical patients. ICG sees different correlation structures and routes them to different historical cohorts for treatment effect estimation.

10 The Covariance Manifold

10.1 Covariance as State

For each unit i (patient, student, customer), I observe multiple measurements over time or across variables: $\mathbf{X}_i = (X_{i1}, X_{i2}, \dots, X_{ip})^\top$ with n_i observations. The **sample covariance matrix** is:

$$\Sigma_i = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)(\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)^\top \quad (12)$$

These covariance matrices live on the **manifold of symmetric positive definite (SPD) matrices**, denoted $\mathcal{S}_p^+$. This is a Riemannian manifold with a natural geometric structure that respects the positive-definiteness constraint.

10.2 Riemannian Distance: Capturing Structure

The standard Euclidean distance between covariance matrices is geometrically inappropriate—it can violate positive-definiteness constraints and distort the natural geometry of the space.

Instead, I use the **Log-Euclidean distance**:

$$d_{LE}(\Sigma_i, \Sigma_j) = \|\log(\Sigma_i) - \log(\Sigma_j)\|_F \quad (13)$$

where $\log(\cdot)$ is the matrix logarithm and $\|\cdot\|_F$ is the Frobenius norm.

This distance has several desirable properties:

- It respects the manifold structure of SPD matrices.
- It is invariant to affine transformations of the data.
- It captures *shape* (correlation structure) without being dominated by *scale* (variance magnitude).

10.3 Mathematical Details

For symmetric positive definite matrices $A, B \in \mathcal{S}_p^+$, the matrix logarithm is computed via eigen-decomposition:

$$\log(A) = V \cdot \text{diag}(\log \lambda_1, \dots, \log \lambda_p) \cdot V^\top \quad (14)$$

for $A = V \cdot \text{diag}(\lambda_1, \dots, \lambda_p) \cdot V^\top$.

The Log-Euclidean metric makes $\mathcal{S}_p^+$ a flat Riemannian manifold (a Lie group), enabling efficient computation of means and distances.

A Note on Computational Cost: While Riemannian computations involve eigendecompositions which can be computationally intensive, there is significant potential for **vectorization** to offset this notable cost. By operating on batches of covariance matrices simultaneously, modern linear algebra libraries can parallelize these operations, making the approach feasible for large datasets.

11 The Unified Distance: Levels + Structure

11.1 Two Sources of Heterogeneity

Treatment effect heterogeneity can be encoded in two distinct ways:

1. **Levels:** Where the unit is located in covariate space. A patient with high baseline inflammation may respond differently than one with low inflammation.
2. **Structure:** How the unit’s measurements covary. A patient whose biomarkers move together may respond differently than one whose biomarkers are independent.

A purely structural distance (Riemannian on covariances) captures correlation patterns but ignores location. A purely level-based distance (Euclidean on means) captures location but ignores structure.

11.2 The Unified ICG Distance

The **Unified ICG** approach combines both:

$$d_{ICG}(\text{Unit}_i, \text{Unit}_j) = d_{LE}(\Sigma_i, \Sigma_j) + \|\mu_i - \mu_j\|_2 \quad (15)$$

where $\mu_i = \mathbb{E}[\mathbf{X}_i]$ is the mean vector of unit i ’s observations.

This unified distance captures both:

- **Levels:** Via Euclidean distance on means
- **Structure:** Via Riemannian distance on covariances

11.3 Just-in-Time Local Estimation

Given a query unit i for whom I wish to estimate the treatment effect, ICG proceeds as follows:

Algorithm 1 Intrinsic Causal Geometry (ICG)

Require: Training data $\{(\mathbf{X}_j, T_j, Y_j)\}_{j=1}^n$, query unit $(\mathbf{X}_i, T_i)$, neighbors k

- 1: Compute mean μ_i and covariance Σ_i from $\mathbf{X}_i$
 - 2: Compute distances $d_{ICG}(\text{Unit}_i, \text{Unit}_j)$ for all training units j
 - 3: Select k nearest neighbors $\mathcal{N}_i = \{j : j \in \text{top-}k \text{ by } d_{ICG}\}$
 - 4: Fit local model on $\{(\mathbf{X}_j, T_j, Y_j) : j \in \mathcal{N}_i\}$
 - 5: Estimate $\hat{\tau}_i = \hat{f}(X_i, T = 1) - \hat{f}(X_i, T = 0)$
 - 6: **return** $\hat{\tau}_i$
-

The critical feature is that matching occurs at *prediction time*, not training time. This “just-in-time” approach avoids locking in static subgroup definitions.

Part IV

Connections to Physics and Invariance

Note on Complexity: The framework presented here introduces significant complexity compared to standard scalar treatment effect models. I argue that this additional complexity is not merely an aesthetic choice but a structural necessity. The connection to physics discussed below serves as a **defence** to this complexity: just as physics requires complex formalisms (like quantum mechanics) to model systems where measurement affects state, causal inference in adaptive systems requires geometric formalisms to capture how treatment (measurement) transforms the system’s structural state.

12 The Hierarchy of Mechanism Stability

12.1 What Makes a Mechanism Invariant?

Not all causal mechanisms are equally stable. I propose a hierarchy:

1. **Physical Laws (Maximally Invariant):** $F = ma$, $E = mc^2$, Maxwell’s equations. These hold regardless of who measures them, when, or why. They do not adapt to observation.
2. **Chemical Mechanisms (Highly Stable):** Enzyme kinetics, reaction rates, binding affinities. These are stable within conditions but can vary with temperature, pH, etc.
3. **Biological Mechanisms (Moderately Stable):** Drug metabolism pathways, immune responses. These vary across individuals and adapt over time (e.g., resistance).
4. **Medical Practice (Moderately Stable, Gradual Decay):** Clinical medicine sits between biology and behavior. The underlying pharmacology is stable (a drug’s mechanism doesn’t change because I discovered it), but deployment is subject to gradual Lucas decay as physicians adjust prescribing patterns and patients modify adherence. The decay rate is slower than in economics—a drug that worked in 1990 likely still works in 2025, even if the observed effect size has shifted.
5. **Behavioral Mechanisms (Unstable):** Economic relationships, social dynamics, treatment adherence. These adapt rapidly to knowledge, incentives, and context.
6. **Reflexive Mechanisms (Maximally Unstable):** Any mechanism that changes rapidly when knowledge of it is disseminated. This includes financial markets, policy responses, and social interventions where agents actively optimize against published findings.

12.2 Observer-Independence in Physical vs. Adaptive Systems

Physical systems exhibit invariance because their objects of study are *non-adaptive*. An electron does not “learn” that physicists expect it to have a certain mass. A photon does not “respond” to predictions about its behavior.

Causal inference in adaptive systems operates differently. The act of causal discovery can change the system being studied:

- **Patients** learn about treatment effects and adjust expectations, adherence, and behavior.

- **Physicians** learn about heterogeneity and adjust prescription patterns.
- **Hospitals** learn about outcomes and adjust protocols.
- **Insurers** learn about cost-effectiveness and adjust coverage.
- **Regulators** learn about safety and adjust requirements.

Each of these adaptations changes the data-generating process, potentially invalidating the very findings that prompted them.

12.3 Random Mutations: The Other Source of Invariance

Beyond physical laws, there is one other source of mechanism invariance: **random mutations in game-theoretic evolution**.

In evolutionary dynamics, random variation generates novel strategies that succeed or fail based on fitness. This process operates independently of observer knowledge:

- A mutation that confers antibiotic resistance will spread regardless of whether microbiologists have characterized it.
- A behavioral strategy that increases reproductive success will be selected for regardless of whether psychologists have named it.

The invariance here comes from the randomness: mutations do not “know” they are being observed. Natural selection operates on fitness, not on scientific publications.

This provides a template for what true causal inference would require: mechanisms that operate independently of knowledge about them. Outside of physics and random mutation, such mechanisms are rare.

12.4 The Adaptive Envelope: Why Medicine Works (And Its Limits)

A fair objection arises: if Lucas’ Critique applies to medicine, why do drugs work at all? Why does aspirin still relieve headaches decades after its mechanism was published?

The answer requires distinguishing between the **chemical law** and its **boundary conditions**:

The chemical law is fixed, but the human agent controls the stoichiometry of the environment.

When aspirin inhibits cyclooxygenase, the reaction $A+B \rightarrow C$ proceeds according to immutable thermodynamics. This is the physics-level invariance that makes medicine possible. The mechanism itself does not adapt to knowledge of it.

But the *density and availability of reagents*—the concentrations of substrates, the timing of administration, the presence of competing molecules—are controlled by adaptive human behavior. Patients decide when to take pills, what to eat, how to live. Physicians decide dosing, timing, and co-prescriptions. These decisions adapt to knowledge.

This creates what I might call the **Chemical Lucas Critique**: the reaction mechanism is invariant, but the reaction *conditions* are set by adaptive agents who respond to causal findings.

12.4.1 The Envelope Interpretation

These examples suggest that medicine operates within an **adaptive envelope**. The inner core—molecular mechanisms—is genuinely invariant. The outer boundary—behavioral conditions that set substrate densities, exposure patterns, and competing pathways—is fully adaptive.

Medical efficacy persists because the core mechanisms are physics-level stable. Medical effect *sizes* drift because the boundary conditions adapt. A drug “works” in the sense that its mechanism is reliable. Whether it achieves its intended population-level effect depends on how human behavior reshapes the environment in which that mechanism operates.

ICG’s relevance here is that covariance structure Σ captures these environmental conditions. Two patients with similar Σ are not just biochemically similar—they exist in similar behavioral environments with similar substrate densities. Matching on geometry matches on the adaptive envelope, not just the invariant core.

13 The Quantum Phase Space Isomorphism

13.1 Mapping ICG to Quantum Mechanics

The Augmented State (μ, Σ) in ICG has a striking isomorphism to quantum phase space representations:

ICG Concept	Quantum Analog	Interpretation
μ (Mean vector)	Displacement	Classical trajectory / Expected position
Σ (Covariance matrix)	Squeezing	Structural geometry / Uncertainty shape
$d_{LE}(\Sigma_i, \Sigma_j)$	Fidelity distance	Similarity of quantum states
Hard classification	Strong measurement	Wave function collapse
Continuous matching	Weak measurement	Trajectory reconstruction

In quantum optics, a *coherent state* is characterized by its displacement (where the state is centered) and its squeezing (the shape of its uncertainty ellipse). The Wigner function representation places quantum states in phase space using exactly these two parameters.

ICG does the same for causal states: the mean μ tells me *where* the unit is (its Level), and the covariance Σ tells me the *shape* of its variability (its Structure).

13.2 Weak Measurement and Trajectory Reconstruction

The analogy extends to methodology. In 2011, Kocsis et al. demonstrated experimentally that **weak measurements** could reconstruct the average trajectories of photons through a double-slit apparatus (Kocsis et al., 2011). Strong measurements collapse the wave function, destroying the interference pattern. Weak measurements extract partial information while preserving coherence, allowing trajectory reconstruction that strong measurements preclude.

ICG operates on the same principle. Traditional methods perform “strong measurement” on latent heterogeneity—forcing units into discrete categories (tree splits, cluster membership, propensity strata). This collapses the continuous variation into categorical membership, destroying information.

ICG performs “weak measurement”—using geometric distance to extract similarity information without forcing categorical collapse. Units retain their position on a continuous manifold. The treatment effect is estimated locally, respecting the uncertainty in latent state rather than eliminating it through hard assignment.

13.3 Why This Matters

The quantum analogy is not merely aesthetic. It suggests that the measurement-disturbance trade-off is *fundamental*—that forcing discrete classifications on continuous phenomena necessarily destroys information. Physics solved this problem with weak measurement. ICG offers an analogous solution for causal inference: preserve structural uncertainty, match continuously, estimate locally.

14 Implications for Causal Inference

14.1 The Scope of Valid Inference

Given this hierarchy, I can characterize where causal inference is on solid ground:

Solid ground:

- Physical processes (drug pharmacokinetics at the molecular level)
- Chemical reactions (binding affinities, metabolic pathways)
- Processes isolated from knowledge dissemination (blinded trials during the trial)

Shaky ground:

- Biological responses that adapt (immune system, microbiome)
- Behavioral outcomes (adherence, lifestyle changes)
- Any effect that depends on patient or provider knowledge

Challenging terrain:

- Economic predictions (markets adapt to forecasts)
- Social interventions (communities adapt to programs)
- Published findings intended to change practice

14.2 Practical Implications

Much of causal inference in medicine, education, and policy operates in domains where mechanisms are not fully invariant—they evolve as knowledge spreads. This is not a counsel of despair; it is a call for appropriate methodology.

Causal inference remains highly valuable in these domains. The practical response is:

1. I should be humble about the shelf-life of causal findings.
2. I should monitor for drift as findings are disseminated.
3. I should build adaptive systems that update as mechanisms change.
4. I should prioritize findings about mechanisms closest to physics.

14.3 The Ergodic Trap: Biological Invariance vs. Systemic Non-Ergodicity

Ergodic theory provides statistical guarantees when time averages equal ensemble averages—when a system’s long-run behavior can be predicted from cross-sectional observations. One might hope that biological mechanisms, being close to physics, provide ergodic guarantees that survive into medical practice.

This hope is misguided. I must distinguish:

- **Biological Ergodicity:** “Chemical X always binds to Receptor Y with affinity K_d .” This may be genuinely time-invariant—a statement about molecular physics that does not change with knowledge.
- **Systemic Non-Ergodicity:** “Humans change how they use Chemical X based on news about Receptor Y.” The deployment of the biological mechanism involves human agents who adapt, learn, and respond to information.

Even if the biology is perfectly stable, the *deployment* of that biology is subject to human agency, which is non-ergodic. Consider a concrete example:

A clinical trial proves Drug A works (invariant biology: the drug genuinely reduces mortality). This finding is published. Physicians, now aware of the drug’s efficacy, begin prescribing it preferentially to high-risk patients—those most likely to die without intervention. In the post-publication observational data, Drug A becomes *associated with higher mortality* (Simpson’s Paradox), because the treated population is systematically sicker than the untreated population.

The causal effect hasn’t changed—the drug still works. But the *observed* causal relationship in the data has flipped, because the human element (physician decision-making) introduced confounding that didn’t exist during the trial.

This is the “Agent-Based Breaker” of statistical guarantees. Ergodic theory assumes the data-generating process is stationary. But when human agents are part of that process, and those agents respond to causal findings, stationarity is violated. The time-invariance that ergodic theory requires is broken by the very act of learning.

The implication is severe: even proving biological invariance (through rigorous RCTs) does not guarantee that causal findings will transport to deployment contexts, because deployment involves human agents who adapt to the findings. The biological mechanism is ergodic; the sociotechnical system embedding it is not.

Part V

Lucas’ Critique Extended

15 The Original Critique

15.1 Lucas in Economics

The economist Robert Lucas argued that econometric relationships estimated from historical data cannot be used reliably for policy evaluation, because the act of implementing a policy changes the system being modeled (Lucas, 1976).

The classic example: if the central bank discovers a relationship between money supply and unemployment (the Phillips curve), and implements policy based on this relationship, economic agents will adapt their expectations, and the relationship will break down.

15.2 Extension to Causal Inference

I extend this critique to causal inference more broadly. The critical insight is that **counterfactual reasoning assumes the treatment is not known to the patient as an option**. The potential outcome $Y(0)$ —the outcome under no treatment—implicitly assumes a world where the patient does not know the treatment exists, has no expectation of receiving it, and experiences no psychological or social effects from its availability.

Once a treatment is known to exist and be beneficial:

A clinical trial shows that treatment T benefits patients compared to no treatment. The moment this finding is published, “no treatment” ceases to exist as a meaningful category. What exists instead is “treatment withheld”—a fundamentally different intervention that occurs in a world where the treatment’s benefits are known.

This is not merely a semantic distinction. “Untreated” and “treatment withheld” are *different interventions* with different causal effects:

- **Untreated:** The patient exists in a world without the treatment. No deprivation is experienced because nothing is being withheld. Baseline care is provided without guilt or compensation.
- **Treatment withheld:** The patient exists in a world where the treatment is known to be beneficial but is not provided. Deprivation is real. Providers may compensate with extra attention. Trust in the system may erode. Nocebo effects may emerge.

16 Gradual Decay, Not Discrete Collapse

16.1 The Decay Framework

A critical refinement: Lucas’ Critique does not operate as a discrete switch. Causal relationships do not suddenly break—they *decay*.

The moment a finding is published, its validity begins to erode. Early adopters may see effects close to the trial estimate. But as knowledge diffuses through the system—through media coverage, physician training, patient advocacy groups, insurance policies—the gap between trial conditions and deployment conditions widens progressively.

16.2 The Half-Life of Causal Estimates

This has implications for how I think about “transportability”:

- There is no binary “transportable vs. non-transportable.”
- Instead, there is a *decay function* that maps time-since-publication (and breadth-of-dissemination) to validity degradation.
- The half-life of a causal estimate depends on how actionable the finding is—findings that change behavior decay faster than findings that merely describe.

We might formalize this as:

$$\text{Validity}(t) = \text{Validity}_0 \cdot e^{-\lambda \cdot D(t)} \quad (16)$$

where $D(t)$ is a measure of dissemination at time t and λ captures how sensitive the mechanism is to knowledge.

For physical mechanisms, $\lambda \approx 0$ —the validity does not decay. For behavioral mechanisms, λ may be large, and validity may decay rapidly once findings become widely known.

16.3 The Non-Ergodic Foundation of Decay

The half-life concept exposes a fundamental connection to **ergodic theory**. Ergodic systems have the property that time averages equal ensemble averages—what I observe over time from one trajectory equals what I would observe across many trajectories at one time. This property underlies most of statistical inference.

But Lucas decay breaks ergodicity. The data-generating process at time t is not the same as the process at time $t + \Delta t$ because knowledge has accumulated and agents have adapted. Time averages no longer equal ensemble averages because the ensemble itself is drifting.

This is why biological invariance does not save me. The underlying pharmacology may be ergodic—a drug binds to receptors the same way in 2025 as in 1995. But the *deployment* of that drug involves human agents: physicians who learn prescribing patterns, patients who form expectations, systems that adapt protocols. The sociotechnical system embedding the biology is non-ergodic.

A drug that “works” in an RCT (biological ergodicity) may show diminishing observed effects over time (systemic non-ergodicity) as the human elements of deployment adapt to knowledge of its efficacy. The biological mechanism is invariant; the causal estimate is not.

16.4 The Counterfactual Corruption

Once a treatment is known, the counterfactual is corrupted:

“Untreated” does not exist once treatment is known. The counterfactual becomes “treatment withheld”—an active intervention that generates placebo effects, trust erosion, and behavioral adaptation. This is not a semantic distinction; it is a causal distinction with measurable consequences.

The “untreated” patient in a pre-publication trial has no expectations—they exist in a world where the treatment’s efficacy is unknown. The “treatment withheld” patient post-publication knows they are being denied something beneficial. Their psychological state, adherence to other treatments, relationship with providers, and physiological stress responses are all affected.

“Untreated” and “treatment withheld” are *different interventions* producing *different outcomes*. The causal effect estimated in the trial—comparing treatment to untreated—does not estimate the effect of treatment vs. withholding in the deployed world.

17 The Recursive Critique

17.1 The Infinite Regress

The situation is worse than a single decay process. Consider what happens when I respond to Lucas’ Critique:

1. An initial SCM identifies causal drivers $\{X_1, \dots, X_k\}$ affecting outcome Y .
2. This finding is published and disseminated.
3. Agents adapt their behavior based on $\{X_1, \dots, X_k\}$, invalidating the original SCM.
4. Researchers identify *new* causal drivers $\{Z_1, \dots, Z_m\}$ that account for the adaptation.
5. This *updated* SCM is published and disseminated.
6. Agents adapt to the *new* drivers, invalidating the updated SCM.
7. The cycle continues indefinitely.

This is the **Recursive Lucas Critique**: every attempt to model the system changes the system, requiring a new model, which changes the system again. There is no fixed point. Causal inference becomes a moving target that recedes as I approach it.

17.2 Game-Theoretic Interpretation

In game-theoretic terms, I am playing against an adversary (the collective behavior of adaptive agents) who observes my strategy (published findings) and best-responds. There is no Nash equilibrium where my causal model remains valid after publication.

The only escape would be to keep findings secret—but this defeats the purpose of science. Or to study systems that cannot adapt—but these are limited to physics and random mutation.

18 The Consequences in Medicine

18.1 How Publication Changes Everything

Consider a clinical trial demonstrating that Drug A reduces mortality. The moment this finding is published:

1. **Patient expectations change:** Patients who receive Drug A expect to benefit, potentially generating placebo effects. Patients denied Drug A expect to do worse, generating nocebo effects. The “untreated” group in the trial—who had no expectation because the drug’s benefit was unknown—no longer exists as a category.
2. **Physician behavior changes:** Doctors who prescribe Drug A do so believing it helps. Doctors who withhold it may feel guilty, communicate differently, or compensate with other interventions. The physicians in the trial—who were in equipoise—no longer exist.
3. **System effects emerge:** Hospitals stock Drug A, train staff in its administration, adjust protocols for its side effects. The healthcare system reorganizes around the drug’s existence. The “baseline care” against which the drug was tested has shifted.
4. **The comparison group transforms:** What was “untreated” becomes “treatment withheld.” These are fundamentally different conditions with different psychological, social, and physiological implications.

18.2 The Estimated Effect Is Not the Deployed Effect

The estimated treatment effect $\hat{\tau}$ from the trial does not estimate the effect of deploying the drug in the real world. It estimates the effect in a hypothetical world where knowledge of the drug’s benefit does not exist. That world vanishes the moment the trial is published.

19 Treatment as Matrix, Not Scalar

19.1 The Knowledge-Inclusive Treatment

A critical insight: **once a treatment is known, the intervention is not just the drug—it is the drug plus the knowledge of the drug.** The moment a finding is published, “treatment” becomes a composite object that includes:

- The pharmacological intervention itself
- Patient expectations about efficacy
- Physician confidence in prescribing
- System-level adaptations (protocols, monitoring, dosing guidelines)
- Social knowledge (what patients tell each other, media coverage)

This composite cannot be captured by a scalar $T \in \{0,1\}$. The “same” drug administered before and after publication of efficacy data is *not the same treatment*.

19.2 Scalar vs. Matrix Effects on the Causal State

The distinction between scalar and matrix treatment maps directly onto my geometric framework:

Scalar Treatment ($+\tau$): A scalar treatment effect shifts the *mean* of the outcome distribution:

$$\mu_{\text{post}} = \mu_{\text{pre}} + \tau \quad (17)$$

This is what traditional frameworks assume—treatment adds a constant (possibly heterogeneous) shift to expected outcomes.

Matrix Treatment (M): A matrix treatment *rotates* or *transforms* the covariance structure:

$$\Sigma_{\text{post}} = M\Sigma_{\text{pre}}M^T \quad (18)$$

This captures treatments that change *relationships* between variables, not just their levels.

19.3 The Decoupling Example

Consider an antihypertensive drug. The traditional question asks: “Does this drug lower blood pressure (shift μ)?” But some drugs may have a more subtle effect:

*The drug may not lower average blood pressure at all—but it may **decouple** blood pressure from stress response.*

Before treatment, blood pressure and cortisol are highly correlated ($\rho = 0.8$). After treatment, they become independent ($\rho = 0$). The *mean* blood pressure is unchanged, but the *covariance structure* has been rotated.

This is a matrix effect: $\Sigma_{\text{post}} \neq \Sigma_{\text{pre}}$, even though $\mu_{\text{post}} = \mu_{\text{pre}}$. A scalar treatment model would conclude “no effect.” A matrix treatment model would detect the structural transformation.

19.4 Emergent Properties of Combined Treatment

The non-additivity problem compounds the matrix issue. Consider a patient receiving Drug A, Drug B, physical therapy, and psychological support. The combined effect is **not the sum of individual effects**:

$$\tau(\mathbf{T}) \neq \tau(A) + \tau(B) + \tau(P) + \tau(S) \quad (19)$$

Drug interactions, synergies, and antagonisms create emergent properties. The *sequence* matters (chemotherapy before vs. after surgery). The *timing* matters (morning vs. evening). The *combination* matters (Drug A alone vs. with Drug B).

19.5 Implications for Causal Inference

Treatment should be modeled as a matrix operator—capturing not just *what* interventions, but *how they transform the joint distribution* of outcomes and covariates. The current practice of estimating $\tau(T)$ for binary T vastly understates the complexity.

Even if I could overcome all other limitations, the scalar treatment assumption would limit applicability. True treatment effects operate on the full state space (μ, Σ) , not just the mean.

Part VI

Empirical Validation

20 The Stress Test: Experimental Design

20.1 Motivation

To validate the ICG framework and expose the failure modes of different approaches, I designed three stress tests:

1. **Noise Swamp**: Signal-to-noise ratio ≈ 0.1 . The treatment effect depends on a signal buried in $5\times$ noise variance. *Requires*: Level matching to lock onto mean signal.
2. **Sinusoidal Trap**: Effect $\tau = 2\sin(x)$, a non-linear function of covariate position. *Requires*: Level matching to trace the manifold locally.
3. **Interaction Gate**: Effect determined by *correlation structure*—coupled variables ($\rho = 0.9$) yield $\tau = 2$, decoupled variables ($\rho = 0$) yield $\tau = 0$. Means and variances are identical across groups. *Requires*: Structure matching to detect covariance shift.

20.2 The Hypothesis

- **Pure Geometry** (Riemannian distance only) should fail on Tests 1 & 2 (where heterogeneity is encoded in means) but succeed on Test 3 (where heterogeneity is encoded in structure).
- **Random Forest** should succeed on Tests 1 & 2 (where standard features capture heterogeneity) but fail on Test 3 (where summary statistics are identical).
- **Unified ICG** (Levels + Structure) should succeed on all three by combining both sources of information.

21 Results

21.1 Main Results

Table 2: Unified Causal Geometry Benchmark: MAE ($\pm$ SD) by Test and Model

Model	Noise Swamp	Sinusoidal	Interaction	TOTAL
Global Linear	0.38 ± 0.48	0.81 ± 0.93	1.00 ± 1.00	2.20
Random Forest	0.11 ± 0.34	0.02 ± 0.02	0.67 ± 0.78	0.80
JIT (Pure Geo)	0.99 ± 1.00	1.28 ± 1.40	0.01 ± 0.01	2.29
JIT (Augmented+Ridge)	0.15 ± 0.33	0.06 ± 0.09	0.01 ± 0.01	0.22
JIT (Augmented+Lasso)	0.16 ± 0.33	0.06 ± 0.09	0.01 ± 0.01	0.23

21.2 The Double Dissociation

The results reveal a **double dissociation**—a pattern where two methods fail on complementary tasks, exposing their orthogonal blind spots:

Random Forest Limitations (Blind to Structure): Random Forest achieves excellent performance on Tests 1 and 2 (MAE = 0.11, 0.02) where heterogeneity is encoded in Levels (μ). But it performs poorly on the Interaction Gate (MAE $\approx$ 0.67). Why? Because Random Forest sees only summary statistics—means and variances—which are *identical* across coupled and decoupled regimes. It is fundamentally blind to covariance structure (Σ). The Interaction Gate exposes this blindness with surgical precision.

Pure Geometry Limitations (Blind to Levels): JIT (Pure Geo) performs poorly on Tests 1 and 2 (MAE $>$ 0.99) because these tests encode heterogeneity in covariate *means*, not *structure*. The Riemannian distance on covariances ignores mean information entirely—two units with identical covariance but opposite mean positions are judged “nearby.” Pure Geometry is blind to Levels (μ).

Unified ICG: Spatially Complete: The Augmented ICG approach succeeds across all three tests because it is **spatially complete**—it spans both the Level space (Euclidean distance on μ) and the Structure space (Riemannian distance on Σ). No heterogeneity escapes detection because no dimension of the causal state space is ignored.

The Augmented models achieve:

- 85% improvement over Pure Geo on Noise Swamp (by adding Level matching)
- 95% improvement over Pure Geo on Sinusoidal Trap (by adding Level matching)
- Maintained near-perfect performance on Interaction Gate (Structure matching preserved)
- **72.5% total improvement** over Random Forest (0.22 vs. 0.80)

The double dissociation proves that *neither* Levels nor Structure alone is sufficient. Only their union provides spatially complete coverage of the causal manifold. The scatter plots in Figure 1 visualize this dissociation: Random Forest predictions cluster around the mean on Test 3 (flat line), while Pure Geometry predictions scatter randomly on Tests 1 and 2 (no correlation with truth).

A critical caveat on generalizability: ICG succeeded on the Interaction Gate because the heterogeneity was *second-order*—encoded in correlation structure ($\rho = 0.9$ vs. $\rho = 0$). Had the heterogeneity been driven by *third-order* structure (e.g., treatment works only for negatively skewed

distributions) or *fourth-order* structure (e.g., treatment works only for leptokurtic distributions), ICG would have failed just as Random Forest failed on covariance structure. This highlights the **hierarchy of geometric approximation**: each method captures heterogeneity up to its representational capacity, and is blind to structure beyond it. ICG extends this capacity from first-order (means) to second-order (covariances), but the frontier of blindness merely shifts—it does not disappear.

21.3 Hypothesis Validation

Test	Pure Geo MAE	Augmented MAE	Improvement
Noise Swamp (Levels)	0.9948	0.1533	84.6%
Sinusoidal (Levels)	1.2848	0.0586	95.4%
Interaction Gate (Structure)	0.0072	0.0069	—

The hypothesis is confirmed: Pure Geometry fails when heterogeneity is in levels; Random Forest fails when heterogeneity is in structure; Unified ICG handles both.

21.4 Model Agnosticism

Ridge and Lasso local models produce nearly identical results, confirming that ICG functions as a valid *routing layer*—directing queries to appropriate historical neighbors—independent of the downstream estimator.

22 Stress Test II: Structural Regimes

22.1 Motivation

While the initial stress tests isolated specific signal types (noise, non-linearity, interaction), real-world data often arrives from mixed causal regimes where the structural relationship between covariates, treatment, and outcome varies. To test resilience against structural shifts, I designed a second stress test comprising three distinct regimes:

1. **Regime A (Clean)**: A randomized control trial setting where treatment assignment is independent of covariates ($T \perp X$). *Ideal for*: JIT-Aug (Geometric Matching).
2. **Regime B (Confounded)**: A classic observational setting where a confounder affects both treatment and outcome ($X \rightarrow T, X \rightarrow Y$). *Ideal for*: X-Learner (Propensity-based adjustment).
3. **Regime C (Collider)**: A selection bias setting where sampling conditions on a collider ($T \rightarrow S \leftarrow Y$). *Ideal for*: X-Learner (Robustness to selection).

22.2 The Hybrid Hypothesis

The hypothesis is that no single model will dominate across all regimes. Geometric matching (JIT-Aug) should excel in clean regimes where structure is informative, while propensity-based meta-learners (X-Learner) should handle confounding and selection bias better. A **distance-blended** approach—using geometric distance to switch between models—should offer the best overall performance.

22.3 Results

Table 3: Structural Regime Stress Test: MAE ($\pm$ SD) by Regime

Model	Regime A	Regime B	Regime C	Avg MAE
JIT-Aug (Baseline)	0.003 $\pm$ 0.004	0.981 $\pm$ 0.004	0.520 $\pm$ 0.011	0.502
X-Learner (Baseline)	0.841 $\pm$ 0.018	0.292 $\pm$ 0.032	0.079 $\pm$ 0.020	0.404
Blend-Threshold (P25)	0.182 $\pm$ 0.347	0.292 $\pm$ 0.032	0.079 $\pm$ 0.020	0.184

22.4 Wins and Limitations

Wins:

- **Regime Specialization:** The test confirmed the "No Free Lunch" theorem in causal inference. JIT-Aug achieved near-perfect recovery in the clean regime (MAE 0.003) but failed under confounding. X-Learner handled confounding and colliders significantly better but struggled in the clean regime.
- **Successful Blending:** The blended model (Blend-Threshold P25) successfully identified the appropriate strategy for each regime, achieving a **63.3% improvement** over the JIT-Aug baseline (MAE 0.184 vs 0.502). It effectively matched the best performing model in Regimes B and C.

Limitations:

- **Imperfect Switching:** While the blended model improved on the average, it did not perfectly recover the performance of JIT-Aug in Regime A (0.182 vs 0.003). The distance metric used for blending introduces some noise, leading to sub-optimal model selection in borderline cases.
- **High Variance:** The standard deviation for the blended model in Regime A is high ($\pm$ 0.347), suggesting that the switching mechanism is occasionally unstable.

22.5 Areas for Further Exploration

The results suggest that **dynamic model selection** based on local geometric structure is a promising path forward. Future work should focus on:

- **Refined Distance Metrics:** Developing distance metrics that specifically detect "confounding signatures" to trigger propensity adjustments only when necessary.
- **Smooth Blending:** Moving from hard thresholding to smooth sigmoid-based weighting to reduce variance in the transition zones between regimes.

Part VII

What ICG Actually Provides

23 Bounded Error Through Continuous Matching

ICG’s primary contribution is replacing *membership error* with *distance error*.

Traditional methods partition units into discrete groups (via tree splits, cluster assignment, or propensity score strata). Units near boundaries are assigned to one group or another, with discontinuous consequences.

ICG uses continuous distances to weight historical information. A unit that is “between” two types receives a blend of information from both, proportional to geometric proximity. Being slightly wrong about the unit’s position produces proportionally small error.

This does not solve the fundamental problems—it does not make invariance testable, does not escape Lucas’ Critique, does not model latent complexity. But it provides **graceful degradation** when my assumptions are wrong.

24 Structure Detection Without Structure Modeling

ICG detects that two units have different correlation structures without attempting to model *why* they differ. This is crucial.

In the Interaction Gate test, Random Forest cannot distinguish units because their summary statistics are identical. ICG distinguishes them because their covariance matrices differ. But ICG does not claim to know *why* one unit has $\rho = 0.9$ and another has $\rho = 0$ —it simply uses this difference for matching.

This represents a philosophical shift: from “understanding causes” to “finding similar cases.” I give up the ambition of causal mechanism discovery in exchange for the humbler goal of principled similarity matching.

25 Explicit Uncertainty About What I Don’t Know

ICG’s distance metric makes explicit what traditional methods hide: the degree to which a query unit resembles historical data.

When a query unit is far from all training units in the unified distance space, ICG can flag this case as requiring human judgment. This triage capability does not solve the problem of unknown unknowns, but it makes them **visible** rather than hidden behind confident predictions.

26 The Fundamental Limits Remain

26.1 What ICG Cannot Do

ICG does not solve the fundamental problems:

1. **Transportability:** ICG cannot verify that mechanisms will hold in deployment contexts.
2. **Lucas’ Critique:** ICG cannot escape the reflexivity of knowledge.

3. **Latent Complexity:** ICG cannot explicitly model the hundred dimensions of latent confounding—but it can *match* units with similar latent signatures through their geometric structure.
4. **Interaction-Only Confounders:** ICG cannot detect confounders that affect only treatment response without baseline signature.
5. **Higher-Order Heterogeneity:** ICG operates on second-order structure (covariance). If treatment effects depend on third-order moments (skewness) or fourth-order moments (kurtosis)—for example, if a drug works only for patients with negatively skewed biomarker distributions—ICG would fail to detect this heterogeneity. The covariance matrix Σ is a second-order proxy, not a complete characterization of distributional shape. Extension to higher-order tensors (coskewness, cokurtosis) would be required to capture such effects.

26.2 What ICG Does Provide

1. **Bounded Error:** Continuous matching produces proportional, not discontinuous, harm.
2. **Structure Detection:** Correlation patterns are used without modeling their causes.
3. **Visible Uncertainty:** Distance from training data is explicit and can trigger triage.
4. **Model Agnosticism:** The routing layer works with any downstream estimator.

27 Implications for Practice

27.1 Abandon the Search for Latent Environments

The concept of “discovering” discrete latent environments should be abandoned. It is epistemologically untenable and empirically unnecessary.

What practitioners should do instead:

- Use geometric matching to find similar historical cases.
- Estimate treatment effects locally within geometrically similar neighborhoods.
- Accept that I am doing sophisticated *interpolation*, not causal *discovery*.

27.2 Acknowledge the Reflexivity of Knowledge

Every published finding changes the world it describes. Practitioners should:

- Recognize that treatment effects estimated from trials apply to a world that no longer exists once the trial is published.
- Monitor for drift between trial conditions and deployment conditions.
- Build adaptive systems that update as the world responds to knowledge.

27.3 Embrace Epistemic Humility

The complexity of latent confounding exceeds my ability to model it. Practitioners should:

- Avoid overconfident causal claims.
- Use methods that degrade gracefully when assumptions fail.
- Prefer continuous matching over discrete classification.
- Build triage systems that flag cases beyond the method’s competence.

27.4 Coupled Interventions: The Killer Feature

Beyond stabilization, ICG enables a class of interventions that are currently mathematically impossible under standard frameworks: **coupled interventions** that operate in two stages on the full state space (μ, Σ) .

27.4.1 The Limitation of Current Methods

Standard causal inference treats lifestyle factors—diet, exercise, sleep, stress—as *confounders to be controlled*. We estimate τ conditional on these variables, holding them fixed. But this framing misses the opportunity to *manipulate* these factors as part of the intervention itself.

The reason is mathematical: standard methods model treatment as a scalar T that shifts μ . They have no representation for interventions that change Σ —the covariance structure itself. Moreover, attempting to model these with two separate models independently is infeasible, as noted by extensions of the Lucas critique: the interaction between the two interventions creates a new state that neither independent model can predict.

27.4.2 ICG’s Matrix Operator Framework

ICG treats lifestyle interventions as **matrix operators** that transform the patient’s covariance structure:

$$\Sigma_{\text{post-lifestyle}} = M_{\text{lifestyle}} \Sigma_{\text{pre}} M_{\text{lifestyle}}^T \quad (20)$$

This opens the door to **two-stage coupled interventions**:

Stage 1 (Rotation): Modify diet, exercise, or lifestyle to “rotate” the patient’s covariance structure into a receptive regime. For example, a proposal to encourage a patient to **lose weight and gain muscle mass** to see if the chances of success for a primary treatment can be improved. This is not controlling for confounders—it is actively *transforming* the patient’s geometric position on the causal manifold.

Stage 2 (Vector): Administer the chemical treatment once the pathway is geometrically open. The drug now operates in a covariance environment optimized for its mechanism.

27.4.3 A Concrete Example

Consider a patient with hypertension where blood pressure is tightly coupled to stress ($\rho = 0.8$). Standard treatment administers an antihypertensive and hopes it overcomes the stress-driven variability.

A coupled intervention proceeds differently:

1. **Stage 1:** Prescribe a stress-reduction protocol (meditation, exercise) that *decouples* blood pressure from cortisol, rotating Σ so that $\rho \rightarrow 0$.
2. **Stage 2:** Administer the antihypertensive once the covariance structure no longer fights the drug’s mechanism.

The drug’s biochemistry is unchanged. But its *effective environment*—the covariance structure in which it operates—has been prepared by the first-stage intervention.

27.4.4 Why This Matters

Coupled interventions are currently impossible to design or evaluate because:

- Standard methods cannot represent Σ -transformations.
- Lifestyle factors are treated as nuisance variables, not intervention targets.
- No framework exists for sequencing covariance rotations with mean shifts.

ICG provides this framework. With longitudinal data, I can:

1. Identify patients whose Σ places them in non-receptive regimes.
2. Design Stage 1 interventions (lifestyle modifications) that rotate Σ toward receptivity.
3. Estimate Stage 2 treatment effects conditional on the transformed geometry.
4. Optimize the full two-stage protocol for individual patients.

27.4.5 Resolving the Matrix Paradox: Closed-Loop Protocols

A careful reader may object: *If I argue that causal mechanisms decay under Lucas’ Critique, how can I propose “Diet $\rightarrow$ Rotation” as if it were a stable law?* This is the Matrix Paradox—proposing interventions while acknowledging that intervention effects are unstable.

The resolution is that coupled interventions should be implemented as **closed-loop protocols**, not static prescriptions:

1. **I do not assume** that “Diet $\rightarrow$ Rotation” is invariant. The relationship between lifestyle modification and covariance transformation is itself subject to Lucas decay, individual variation, and contextual factors.
2. **Instead, I verify.** After applying Stage 1 (the lifestyle intervention), I *re-measure* the patient’s covariance state Σ_{t+1} .
3. **Only if** the patient has successfully “rotated” into the receptive geometric regime—verified by measurement, not assumed from mechanism—do I administer Stage 2 (the chemical treatment).
4. **If rotation fails**, I iterate: adjust the lifestyle protocol, re-measure, and verify again before proceeding.

This removes the reliance on “trusting” that the Stage 1 mechanism worked. I rely instead on *verifying the state change*. The closed-loop structure means I am not predicting that diet will rotate Σ ; I am *checking* whether it did.

This is consistent with my broader framework: mechanisms are unstable, but *states are measurable*. I cannot trust that an intervention will produce a specific geometric transformation. I *can* measure whether it did, and condition subsequent actions on that measurement.

The “killer feature” of ICG is not just better matching; it is the ability to reason about interventions that operate on covariance structure itself—implemented as closed-loop protocols that verify state change rather than assuming mechanism stability.

28 Reclaiming Psychology: The JIT Resolution to Non-Ergodicity

Psychology and behavioral sciences face a dual crisis. First, the **replication crisis**, often attributed to p-hacking, is increasingly understood as an **ergodicity crisis**: statistical regularities found in group averages (ensembles) do not mathematically apply to individuals over time unless the system is ergodic. Human behavior is rarely ergodic.

Second, **Lucas’ Critique** doubles down on this instability. Not only do group averages fail to map to individuals, but the very act of establishing an evidence base changes the behavior it categorizes. “Evidence-based practice” that relies on static transportability of past findings to future patients is thus built on a foundation of sand.

ICG provides a specific resolution: **abandon the goal of transportability**.

Instead of asking “Does CBT work for depression?” (a question assuming a stable, transportable average), JIT analysis asks: “For this specific patient, with this specific covariance structure of symptoms, what did the trajectory look like for their geometric neighbors?”

We do not aim to transport a static truth from 1990 to 2025. We aim to find the **physics of the data Just-in-Time**. By treating the patient’s current state (Levels + Structure) as a unique initial condition and matching it to the most relevant historical trajectories, we bypass the need for the system to be ergodic. We don’t need the group average to represent the individual; we only need the local manifold geometry to be locally typically populated.

This shifts the clinical goal from “applying the evidence base” to “interrogating the immediate causal geometry.” It is the only rigorous path for personalized intervention in non-ergodic, reflexive systems.

29 Conclusion

Causal inference has achieved remarkable progress over the past three decades. The development of potential outcomes frameworks, directed acyclic graphs, and machine learning-based heterogeneous treatment effect estimators represents genuine intellectual achievement. Methods like Causal Forests, meta-learners, and double machine learning have moved the field from population averages to individual-level predictions, enabling precision medicine and targeted policy interventions.

This paper does not dismiss these achievements. Rather, it seeks to *extend* the conversation by identifying vulnerabilities that warrant attention—not to abandon causal inference, but to practice it with appropriate caution.

The central contributions are:

1. **Geometric matching complements feature matching**: When treatment effects depend on the structure of covariance among measurements, Riemannian geometry provides addi-

tional matching criteria that feature-based methods may miss. This is not a replacement but an extension.

2. **Continuous distance offers safety margins:** Being slightly wrong about position on a manifold produces bounded, proportional error; misclassifying discrete group membership can produce discontinuous harm. Continuous methods provide graceful degradation.
3. **Certain problems have inherent limits:** Transportability is difficult to verify prospectively, reflexivity affects adaptive systems, and latent complexity exceeds modeling capacity in many practical settings. Acknowledging these limits is not defeatism—it is prudent engineering.
4. **Lucas’ Critique extends beyond economics:** The recursive nature of knowledge dissemination affects any system where agents adapt to published findings. Medicine is more stable than finance, but not immune.
5. **Treatment effects are often high-dimensional:** Combined interventions have emergent properties. The scalar treatment assumption, while useful, may understate complexity in multi-drug, multi-lifestyle intervention regimes.
6. **Mechanism stability varies by domain:** Physical and chemical mechanisms are highly stable; behavioral and reflexive mechanisms are less so. Matching the inferential approach to the mechanism’s stability class improves reliability.

Intrinsic Causal Geometry does not solve the fundamental problems of causal inference. It provides a practical tool that extends current methods to second-order structure and degrades gracefully when assumptions are violated.

A Balanced View

The view expressed here—that latent confounding is often practically impossible to model fully—is driven by experience, not pessimism. In essentially every applied problem I have encountered, I can identify dozens to hundreds of features that plausibly affect treatment response and cannot be measured. This is not a criticism of existing methods; it is an observation about the complexity of real systems.

The appropriate response is not to abandon causal inference but to practice it with humility: acknowledge uncertainty, monitor for drift, use methods that bound rather than hide error, and remain open to revision as systems evolve.

Final Thought

Causal inference in adaptive systems operates under constraints that purely observational or purely experimental disciplines do not face. The methods that acknowledge these constraints—that build in verification, that bound error, that track geometric distance to training data—are likely to prove most durable.

I am not discovering eternal truths about causal structure. I am making pragmatic decisions under irreducible uncertainty, in systems that respond to my decisions. The honest practice of causal inference requires this recognition.

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A Theoretical Results

A.1 Theorem: Orthogonality Impossibility

Theorem 1 (Orthogonality Impossibility). *Let U be a latent confounder affecting treatment effects, and let X be observed covariates such that $U \perp X$ (orthogonality). If $\tau_{\text{lat}}(U)$ is non-constant, then no function $h : \mathcal{X} \rightarrow \mathbb{R}$ can recover $\tau_{\text{lat}}(U_i)$ from X_i alone.*

Proof. Suppose for contradiction that there exists a function $h : \mathcal{X} \rightarrow \mathbb{R}$ such that $h(X_i) = \tau_{\text{lat}}(U_i)$ for all i .

By orthogonality ($U \perp X$), the distribution of U given $X = x$ is independent of x :

$$P(U \mid X = x) = P(U) \quad \text{for all } x \quad (21)$$

Since $\tau_{lat}(U)$ is non-constant, there exist u_1, u_2 such that $\tau_{lat}(u_1) \neq \tau_{lat}(u_2)$.

For any fixed $x \in \mathcal{X}$, both u_1 and u_2 can occur with positive probability given $X = x$ (since $P(U \mid X = x) = P(U)$).

But $h(x)$ is deterministic and returns a single value. Therefore:

$$h(x) \neq \tau_{lat}(u_1) \quad \text{or} \quad h(x) \neq \tau_{lat}(u_2) \quad (22)$$

This contradicts the assumption that $h(X_i) = \tau_{lat}(U_i)$ for all i . Therefore, no such function h exists. $\square$

Implication: Under orthogonality, feature-based methods are **mathematically constrained**—not just difficult, but fundamentally limited.

A.2 Lemma: Residuals Break Orthogonality

Lemma 1. *If U affects baseline outcomes (Assumption 3), then baseline residuals $R = Y_0 - \hat{f}(X)$ are **not** orthogonal to U , even when $U \perp X$.*

Proof. By Assumption 3, $Y_0 = f(X) + g(U) + \epsilon$ where $g(U)$ is non-constant.

The residual is:

$$R = Y_0 - \hat{f}(X) = f(X) + g(U) + \epsilon - \hat{f}(X) = [f(X) - \hat{f}(X)] + g(U) + \epsilon \quad (23)$$

If $\hat{f}$ is trained on (X, Y_0) , then $\mathbb{E}[f(X) - \hat{f}(X) \mid X] \approx 0$ (residuals are uncorrelated with X by construction).

Therefore:

$$R \approx g(U) + \epsilon \quad (24)$$

Since $g(U)$ is non-constant and $\epsilon \perp U$, one has $R \not\perp U$. Thus, residuals contain information about U even when X does not. $\square$

Implication: Residuals provide an “escape hatch” from orthogonality by exploiting baseline coupling.

B Mathematical Details: Log-Euclidean Distance

For symmetric positive definite matrices $A, B \in \mathcal{S}_p^+$, the Log-Euclidean distance is defined as:

$$d_{LE}(A, B) = \|\log(A) - \log(B)\|_F \quad (25)$$

The matrix logarithm is computed via eigendecomposition:

$$\log(A) = V \cdot \text{diag}(\log \lambda_1, \dots, \log \lambda_p) \cdot V^\top \quad (26)$$

for $A = V \cdot \text{diag}(\lambda_1, \dots, \lambda_p) \cdot V^\top$.

B.1 Why Riemannian Geometry Captures Structure

Consider two covariance matrices with identical variances but different correlations:

$$\Sigma_1 = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}, \quad \Sigma_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (27)$$

The Euclidean distance (Frobenius norm of difference) is:

$$\|\Sigma_1 - \Sigma_2\|_F = \sqrt{2 \cdot 0.9^2} = 1.27 \quad (28)$$

The Log-Euclidean distance is:

$$d_{LE}(\Sigma_1, \Sigma_2) = \|\log(\Sigma_1) - \log(\Sigma_2)\|_F \approx 2.08 \quad (29)$$

The Riemannian distance is larger because it accounts for the curvature of the SPD manifold—the “geometric cost” of transforming one correlation structure into another is greater than the element-wise difference suggests.